Hw3

3. The Maclaurin series for the arctangent function converges for $-1 < x \leq 1$ and is given by

$$\arctan x = \lim_{n\to\infty} P_n(x) = \lim_{n\to\infty} \sum_{i=1}^{n} (-1)^{i+1} \frac{x^{2i-1}}{2i-1}.$$

a. Use the fact that $\tan \pi/4 = 1$ to determine the number of n terms of the series that need to be summed to ensure that $|4P_n(1) - \pi| < 10^{-3}$.

b. The C++ programming language requires the value of π to be within 10^{-10}. How many terms of the series would we need to sum to obtain this degree of accuracy?

(a) $\arctan(1) = \frac{\pi}{4} = \sum_{i=1}^{\infty} (-1)^{i+1} \frac{1}{2i-1}$

$|4P_n(1) - \pi| < 10^{-3}$

$P_n(1) = \sum_{i=1}^{n} (-1)^{i+1} \frac{1}{2i-1}$

$4P_n(1) - \pi = 4\sum_{i=1}^{n} (-1)^{i+1}\frac{1}{2i-1} - 4\sum_{i=1}^{\infty}(-1)^{i+1}\frac{1}{2i-1} = -4\sum_{i=n+1}^{\infty}(-1)^{i+1}\frac{1}{2i-1}$

$|4P_n(1) - \pi| = 4\left|\sum_{i=n+1}^{\infty}(-1)^{i}\frac{1}{2i-1}\right|$

$\left|\sum_{i=n+1}^{\infty}(-1)^{i+1}\frac{1}{2i-1}\right| \leq \frac{1}{2(n+1)-1} = \frac{1}{2n+1}$

$|4P_n(1) - \pi| \leq 4 \cdot \frac{1}{2(n+1)-1} = \frac{4}{2n+1}$

$\frac{4}{2n+1} < 10^{-3} \Rightarrow 2n+1 > 4000$

$n > 1999.5$

$\therefore n = 2000$

(b) $|4P_n(1) - \pi| \leq \frac{4}{2n+1}$

$\frac{4}{2n+1} < 10^{-10}$

$2n > 4\times 10^{10} - 1$

$n > 2\times 10^{10} - \frac{1}{2}$

$\therefore n_{min} = 2\times 10^{10}$

6. Find the rates of convergence of the following sequences as $n \to \infty$.

a. $\lim_{n\to\infty} \sin\frac{1}{n} = 0$

b. $\lim_{n\to\infty} \sin\frac{1}{n^2} = 0$

c. $\lim_{n\to\infty} \left(\sin\frac{1}{n}\right)^2 = 0$

d. $\lim_{n\to\infty} [\ln(n+1) - \ln(n)] = 0$

7. Find the rates of convergence of the following functions as $h \to 0$

6. a. $\lim_{n\to\infty} \frac{\sin\frac{1}{n}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{-\frac{1}{n^2}\cos\frac{1}{n}}{-\frac{1}{n^2}} = 1$

$\therefore$ rate of convergence: $O(\frac{1}{n})$

b. $\lim_{h\to 0} \frac{\sin(\frac{1}{n^2})}{\frac{1}{n^2}} = \lim_{n\to\infty} \frac{-2n^{-3}\cos(\frac{1}{n^2})}{-2n^{-3}} = 1$

$\therefore$ rate of convergence: $O(\frac{1}{n^2})$

c. $\lim_{h\to 0} \frac{(\sin\frac{1}{n})^2}{(\frac{1}{n})^2} = \lim_{n\to\infty} \left(\frac{\sin\frac{1}{n}}{\frac{1}{n}}\right)^2$

let $x = \frac{1}{n}$ $\lim_{x\to 0} \frac{\sin x}{x} = 1$

$\lim_{x\to 0} (\frac{\sin x}{x})^2 = (\lim_{x\to 0} \frac{\sin x}{x})^2 = 1$

$\therefore (\sin\frac{1}{n})^2 \sim \frac{1}{n^2}$ $n\to\infty$

$\therefore$ rate of convergence $O(\frac{1}{n^2})$

d. $\lim_{n\to\infty} (\ln(n+1) - \ln n) = \lim_{n\to\infty} \left(\ln\frac{n+1}{n}\right) = \lim_{n\to\infty} \ln(1+\frac{1}{n})$

$\lim_{n\to\infty} \frac{\ln(1+\frac{1}{n})}{\frac{1}{n}} = \lim_{n\to\infty} \frac{\frac{1}{n(n+1)}}{-\frac{1}{n^2}} = \lim_{n\to\infty} \frac{n}{n+1} = 1$

$\therefore$ rate of convergence $O(\frac{1}{n})$

7. Find the rates of convergence of the following functions as $h \to 0$.

a. $\lim_{h\to 0} \frac{\sin h}{h} = 1$ b. $\lim_{h\to 0} \frac{1-\cos h}{h} = 0$

c. $\lim_{h\to 0} \frac{\sin h - h\cos h}{h} = 0$ d. $\lim_{h\to 0} \frac{1-e^h}{h} = -1$

7.(a) $\lim_{h\to 0} \frac{\sin h}{h} = 1$

let $f(h) = \frac{\sin h}{h} - 1$

$\because \sin h = h - \frac{h^3}{6} + O(h^5)$

$\therefore \frac{\sin h}{h} - 1 = -\frac{h^2}{6} + O(h^4)$

$\therefore$ rate of convergence : $O(h^2)$

(b) $\lim_{h\to 0} \frac{1-\cos h}{h} = 0$

$f(h) = \frac{1-\cos h}{h}$

$\cos h = 1 - \frac{h^2}{2} + O(h^4)$

$\frac{1-\cos h}{h} = \frac{h}{2} + O(h^3)$

$\therefore$ rate of convergence : $O(h)$

(c) $\lim_{h\to 0} \frac{\sin h - h\cos h}{h} = 0$

$\sin h - h\cos h = (h - \frac{h^3}{6} + O(h^5)) - h(1 - \frac{h^2}{2} + O(h^4)) = \frac{h^3}{3} + O(h^5)$

$\frac{\sin h - h\cos h}{h} = \frac{h^2}{3} + O(h^4)$

$\therefore$ rate of convergence : $O(h^2)$

(d) $\lim_{h\to 0} \frac{1-e^h}{h} = -1$

$f(h) = \frac{1-e^h}{h} + 1$

$e^h = 1 + h + \frac{h^2}{2} + O(h^3)$

$\frac{1-e^h}{h} + 1 = \frac{1-(1+h+\frac{h^2}{2}+O(h^3))}{h} + 1 = -\frac{h}{2} + O(h^2)$

$\therefore$ rate of convergence : $O(h)$

8. Suppose that $0 < q < p$ and that $\alpha_n = \alpha + O(n^{-p})$.

a. Show that $\alpha_n = \alpha + O(n^{-q})$.

b. Make a table listing $1/n$, $1/n^2$, $1/n^3$, and $1/n^4$ for $n = 5, 10, 100$, and 1000 and discuss the varying rates of convergence of these sequences as n becomes large.

a. $\alpha_n = \alpha + O(n^{-p})$

there exist $C > 0$ and $N > 0$, let $n > N$

$|\alpha_n - \alpha| \le C \cdot n^{-p}$

$\because 0 < q < p$

$\therefore n^{-p} \le n^{-q}$ when $n \ge 1$

$\therefore |\alpha_n - \alpha| \le C \cdot n^{-p} \le C \cdot n^{-q}$

let $k = C$, $\forall n > N$

$|\alpha_n - \alpha| \le k \cdot n^{-q}$

$\therefore \alpha_n = \alpha + O(n^{-q})$

b.

n	$\frac{1}{n}$	$\frac{1}{n^2}$	$\frac{1}{n^3}$	$\frac{1}{n^4}$
5	0.2	0.04	0.008	0.0016
10	0.1	0.01	0.001	0.0001
100	0.01	0.0001	0.000001	0.00000001
1000	0.001	0.000001	0.000000001	0.000000000001

Conclusion: The larger the power p is, the faster the sequence $\frac{1}{n^p}$ converges to 0.

10. Suppose that as x approaches zero,

$$F_1(x) = L_1 + O(x^\alpha) \quad \text{and} \quad F_2(x) = L_2 + O(x^\beta).$$

Let c_1 and c_2 be nonzero constants and define

$$F(x) = c_1F_1(x) + c_2F_2(x) \quad \text{and} \quad G(x) = F_1(c_1x) + F_2(c_2x).$$

Show that if γ = minimum $\{\alpha, \beta\}$, then, as x approaches zero,

a. $F(x) = c_1L_1 + c_2L_2 + O(x^\gamma)$

b. $G(x) = L_1 + L_2 + O(x^\gamma)$.

(a) $\gamma = \min\{\alpha, \beta\}$

$F_1(x) = L_1 + O(x^\alpha)$

$F_2(x) = L_2 + O(x^\beta)$

$\exists\ C_1, C_2 > 0$

$|F_1(x) - L_1| \le C_1|x|^\alpha$

$|F_2(x) - L_2| \le C_2|x|^\beta$

$F(x) = c_1F_1(x) + c_2F_2(x)$

$= c_1(L_1 + O(x^\alpha)) + c_2(L_2 + O(x^\beta))$

$= c_1L_1 + c_2L_2 + c_1O(x^\alpha) + c_2O(x^\beta)$

$\because \gamma \le \alpha \ \therefore O(x^\alpha) = O(x^\gamma)$

$\because \gamma \le \beta \ \therefore O(x^\beta) = O(x^\gamma)$

$c_1 \cdot O(x^\gamma) = O(x^\gamma) \quad c_2 \cdot O(x^\gamma) = O(x^\gamma)$

$O(x^\gamma) + O(x^\gamma) = O(x^\gamma)$

$c_1 \cdot O(x^\alpha) + c_2O(x^\beta) = O(x^\gamma) + O(x^\gamma) = O(x^\gamma)$

$\therefore F(x) = c_1L_1 + c_2L_2 + c_1O(x^\alpha) + c_2O(x^\beta) = c_1L_1 + c_2L_2 + O(x^\gamma)$

(b) $G(x) = F_1(c_1x) + F_2(c_2x)$

$F_1(x) = L_1 + O(x^\alpha)$

$F_1(c_1x) = L_1 + O((c_1x)^\alpha) = L_1 + O(|c_1|^\alpha|x|^\alpha) = L_1 + O(x^\alpha)$

$F_2(c_2x) = L_2 + O((c_2x)^\beta) = L_2 + O(|c_2|^\beta|x|^\beta) = L_2 + O(x^\beta)$

$\therefore G(x) = F_1(c_1x) + F_2(c_2x) = L_1 + O(x^\alpha) + L_2 + O(x^\beta)$

let $\gamma = \min\{\alpha, \beta\}$

$O(x^\alpha) + O(x^\beta) = O(x^\gamma)$

$\therefore G(x) = L_1 + L_2 + O(x^\gamma)$

11. The sequence $\{F_n\}$ described by $F_0 = 1$, $F_1 = 1$, and $F_{n+2} = F_n + F_{n+1}$, if $n \ge 0$, is called a *Fibonacci sequence*. Its terms occur naturally in many botanical species, particularly those with petals or scales arranged in the form of a logarithmic spiral. Consider the sequence $\{x_n\}$, where $x_n = F_{n+1}/F_n$. Assuming that $\lim_{n\to\infty} x_n = x$ exists, show that $x = (1 + \sqrt{5})/2$. This number is called the *golden ratio*.

11. $x_n = \frac{F_{n+1}}{F_n}$

$\frac{F_{n+2}}{F_{n+1}} = \frac{F_n}{F_{n+1}} + 1$

$x_{n+1} = \frac{1}{x_n} + 1$

$\lim_{n\to\infty} x_n = x \quad \lim_{n\to\infty} x_{n+1} = x$

$\therefore x = \frac{1}{x} + 1 \quad (x \ne 0)$

$x^2 = 1 + x$

$x^2 - x - 1 = 0$

$\because x > 0$

$\therefore x = \frac{1+\sqrt{5}}{2}$

1. Use the Bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x = 0$ on $[0, 1]$.
2. Let $f(x) = 3(x+1)(x-\frac{1}{2})(x-1) = 0$. Use the Bisection method on the followine intervals to finc

P_1:

1. $a_0=0, b_0=1$

$f(0)=0-1=-1$ $f(1)=1-\cos 1 \approx 0.4597 > 0$ $f(0)\cdot f(1)<0$

$f(\frac{1}{2}) = \sqrt{\frac{1}{2}} - \cos\frac{1}{2} = -0.1705 < 0$

$\because f(\frac{1}{2})\cdot f(1)<0 \therefore a_1=0.5 \quad b_1=1$

P_2: $f(\frac{3}{4}) = \sqrt{\frac{3}{4}} - \cos\frac{3}{4} = 0.1343$

$\because f(\frac{1}{2})\cdot f(\frac{3}{4})<0 \therefore a_2=0.5 \quad b_2=0.75$

P_3: $P_3 = \frac{0.5+0.75}{2} = \frac{5}{8}$

$f(\frac{5}{8}) = \sqrt{\frac{5}{8}} - \cos\frac{5}{8} = -0.0204$

4. Use the Bisection method to find solutions accurate to within 10^{-2} for $x^4 - 2x^3 - 4x^2 + 4x + 4 = 0$ on each interval.
   a. $[-2, -1]$ b. $[0, 2]$ c. $[2, 3]$ d. $[-1, 0]$

a. $f(-2) = (-2)^4 - 2(-2)^3 - 4(-2)^2 + 4(-2) + 4 = 12$

$f(-1) = -1$

$x \approx -1.4141$

b. $f(0) = 4$

$f(2) = -4$

$x \approx 1.4141$

c. $f(2) = -4$

$f(3) = 7$

$x \approx 2.7266$

d. $f(-1) = -1$

$f(0) = 4$

$x \approx -0.7266$

```
function bisection_roots
    f=@(x)x.^4 -2*x.^3 - 4*x.^2 + 4*x + 4;
    intervals ={[-2,-1],[0,2],[2,3],[-1,0]};
    tol = 1e-2;
    max_iter = 100;

    for k = 1:length(intervals)
        a = intervals{k}(1);
        b = intervals{k}(2);
        fprintf('region [%g, %g]:\n', a, b);
        root = bisection(f, a, b, tol, max_iter);
        fprintf('  sol = %.4f\n\n', root);
    end
end
function root = bisection(f, a, b, tol, max_iter)
    fa = f(a);
    fb = f(b);
    if fa * fb > 0
        error('error');
    end

    for i = 1:max_iter
        c = (a + b)/2;
        fc = f(c);

        if abs(fc) < tol || (b - a)/2 < tol
            root = c;
            return;
        end

        if fa * fc < 0
            b = c;
            fb = fc;
        else
            a = c;
            fa = fc;
        end
    end

    root = (a + b)/2;
end
```

```
>> bisection_roots
region [-2, -1]:
  sol = -1.4141

region [0, 2]:
  sol = 1.4141

region [2, 3]:
  sol = 2.7266

region [-1, 0]:
  sol = -0.7266
```

12. Let $f(x) = (x+2)(x+1)^2x(x-1)^3(x-2)$. To which zero of f does the Bisection method converge when applied on the following intervals?

a. $[-1.5, 2.5]$ b. $[-0.5, 2.4]$ c. $[-0.5, 3]$ d. $[-3, -0.5]$

$f(x)=0 \quad x_1=-2 \; x_2=-1 \; x_3=0 \; x_4=1 \; x_5=2$

a. $f(-1.5)<0 \quad f(2.5)>0$ convergence to $x=0$

b $f(-0.5)<0 \quad f(2.4)>0$ convergence to $x=0$

C. $f(-0.5)<0 \quad f(3)>0$ convergence to $x=2$

d. $f(-3)>0 \quad f(-0.5)<0$ convergence to $x=-2$

```
function bisection_roots
    f = @(x) (x+2).*(x+1).^2 .* x .* (x-1).^3 .* (x-2);


    intervals = {[-1.5, 2.5], [-0.5, 2.4], [-0.5, 3], [-3, -0.5]};

    tol = 1e-4;
    max_iter = 100;

    for k = 1:length(intervals)
        a = intervals{k}(1);
        b = intervals{k}(2);
        fprintf('region [%g, %g]:\n', a, b);
        root = bisection(f, a, b, tol, max_iter);
        fprintf('  sol = %.4f\n\n', root);
    end
end
```

```
function root = bisection(f, a, b, tol, max_iter)
    fa = f(a);
    fb = f(b);
    if fa * fb > 0
        error('error');
    end

    for i = 1:max_iter
        c = (a + b)/2;
        fc = f(c);

        if abs(fc) < tol || (b - a)/2 < tol
            root = c;
            return;
        end

         if fa * fc < 0
             b = c;
             fb = fc;
         else
             a = c;
             fa = fc;
         end
     end

     root = (a + b)/2;
end
```

```
>> bisection_roots
region [-1.5, 2.5]:
  sol = 0.0000

region [-0.5, 2.4]:
  sol = -0.0001

region [-0.5, 3]:
  sol = 2.0000

region [-3, -0.5]:
  sol = -2.0000
```

13. Find an approximation to $\sqrt[3]{25}$ correct to within 10^{-4} using the Bisection Algorithm. [*Hint:* Consider $f(x) = x^3 - 25$.]

$2^3 = 8 < 25 \quad f(2) = -17 < 0$

$3^3 = 27 > 25 \quad f(3) = 27 - 25 = 2 > 0$

$\therefore$ let $a = 2,\ b = 3$

ans = 2.924

```
function root = cube_root_25
    f = @(x) x.^3 - 25;
    a = 2;
    b = 3;
    tol = 1e-4;
    max_iter = 100;
    fa = f(a);
    fb = f(b);
    if fa * fb > 0
        error('error');
    end
    for i = 1:max_iter
        c = (a + b)/2;
        fc = f(c);

        if abs(fc) < tol || (b - a)/2 < tol
            root = c;
            fprintf('sol = %.6f\n', root);
            fprintf('iter = %d\n', i);
            return;
        end


       if fa * fc < 0
           b = c; fb = fc;
       else
           a = c; fa = fc;
       end
   end

   root = (a + b)/2;
   fprintf('sol = %.6f (Not fully converged)\n', root);
end
```

```
>> cube_root_25
sol = 2.924011
iter = 14

ans =

    2.9240
```